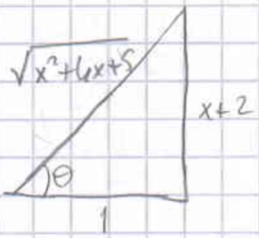
(37) $\int \sqrt{3x^2+12x+15}\,dx$

$x+2=\tan\theta \Rightarrow dx=\sec^2\theta\, d\theta$

$$\int \sqrt{3x^2+12x+15}\,dx = \sqrt{3}\int\sqrt{x^2+4x+5}\,dx = \sqrt{3}\int\sqrt{(x+2)^2+1}\,dx$$

$$= \sqrt{3}\int\sqrt{\tan^2\theta+1}\,.\,\sec^2\theta\, d\theta = \sqrt{3}\int \sec^3\theta\, d\theta$$



$$= \frac{\sqrt{3}}{2}\left[\sec\theta\tan\theta + \ln\left|\sec\theta+\tan\theta\right|\right] + c$$

$$= \frac{\sqrt{3}}{2}\left[\sqrt{x^2+4x+5}\,.\,(x+2) + \ln\left|\sqrt{x^2+4x+5}+x+2\right| + C\right.$$

(38) $\int \frac{5x-3}{x^2-2x-3}\,dx = \int\frac{A\,dx}{x+1} + \int\frac{B\,dx}{x-3}$

$$\left.\begin{array}{l} A+B=5 \\ -3A+B=-3 \end{array}\right\} \Rightarrow A=2 \quad B=3$$

$$\int\frac{5x-3}{x^2-2x-3}\,dx = 2\int\frac{dx}{x+1} + 3\int\frac{dx}{x-3} = 2\ln|x+1| + 3|x-3| + c$$

(39) $\int \frac{(x^2+4x+1)\,dx}{(x-1)(x+1)(x+3)} = A\int\frac{dx}{(x-1)} + B\int\frac{dx}{x+1} + C\int\frac{dx}{x+3}$

$x^2+4x+1 = A(x^2+4x+3) + B(x^2+2x-3) + C(x^2-1)$

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 1 \\ 4 & 2 & 0 & 4 \\ 3 & -3 & -1 & 1 \end{array}\right] \sim \left[\begin{array}{ccc|c} 1 & 1 & 1 & 1 \\ 0 & -2 & -4 & 0 \\ 0 & -6 & -4 & -2 \end{array}\right] \sim \left[\begin{array}{ccc|c} 1 & 0 & -1 & 1 \\ 0 & 1 & 2 & 0 \\ 0 & 0 & 8 & -2 \end{array}\right] \sim \left[\begin{array}{ccc|c} 1 & 0 & 0 & 3/4 \\ 0 & 1 & 0 & 1/2 \\ 0 & 0 & 1 & -1/4 \end{array}\right]$$

$$\int\frac{x^2+4x+1}{(x-1)(x+1)(x+3)}\,dx = \frac{3}{4}\int\frac{dx}{x-1} + \frac{1}{2}\int\frac{dx}{x+1} - \frac{1}{4}\int\frac{1}{x+3}$$

$$= \frac{3}{4}\ln|x-1| + \frac{1}{2}\ln|x+1| - \frac{1}{4}\ln|x+3| + c$$

(40) $\int \frac{6x+7}{(x+2)^2}\,dx = A\int \frac{dx}{x+2} + B\int \frac{dx}{(x+2)^2}$

$6x+7 = A(x+2)+B \Rightarrow A=6 \quad B=-5$

$\int \frac{6x+7}{(x+2)^2}\,dx = 6\int \frac{dx}{x+2} - 5\int \frac{dx}{(x+2)^2} = 6\ln|x+2| + \frac{5}{x+2} + C$

(41) $\int \frac{2x^3-4x^2-x-3}{x^2-2x-3}\,dx$

$$\begin{array}{l|l} 2x^3-4x^2-x-3 & x^2-2x-3 \\ \underline{-2x^3 \pm 4x^2 \pm 6x} & 2x \\ \qquad 5x-3 & \end{array}$$

$\int \frac{2x^3-4x^2-x-3}{x^2-2x-3}\,dx = \int 2x\,dx + \int \frac{5x-3}{x^2-2x-3}\,dx$

$= x^2 + 2\ln|x+1| + 3\ln|x-3| + C$

(42) $\int \frac{-2x+4}{(x^2+1)(x-1)^2}\,dx = \int \frac{Ax+B}{x^2+1}\,dx + C\int \frac{dx}{x-1} + D\int \frac{dx}{(x-1)^2}$

$-2x+4 = (Ax+B)(x^2-2x+1) + C(x^2+1)(x-1) + D(x^2+1)$

$-2x+4 = Ax^3 - 2Ax^2 + Ax + Bx^2 - 2Bx + B + Cx^3 - Cx^2 + Cx - C + Dx^2 + D$

$-2x+4 = (A+C)x^3 + (-2A+B-C+D)x^2 + (A-2B+C)x + (B-C+D)$

$A+C=0 \Rightarrow 2+C=0 \Rightarrow C=-2$

$-2A+\overbrace{B-C+D}^{4}=0 \Rightarrow -2A=4 \Rightarrow A=2$

$A-2B+C=-2 \Rightarrow \cancel{2}-2B\cancel{-2}=\cancel{-2} \Rightarrow B=1$

$B-C+D=4 \Rightarrow 1+2+D=4 \Rightarrow D=1$

$\int \frac{-2x+4}{(x^2+1)(x-1)^2}\,dx = \int \frac{2x+1}{x^2+1}\,dx - 2\int \frac{dx}{x-1} + \int \frac{dx}{(x-1)^2}$

$= \int \frac{2x\,dx}{x^2+1} + \int \frac{dx}{x^2+1} - 2\int \frac{dx}{x-1} + \int \frac{dx}{(x-1)^2}$

$= \ln(x^2+1) + \arctan x - 2\ln|x-1| - \frac{1}{x-1} + C$

(43) $\displaystyle\int \sec x \, dx$

$\sin x = u \Rightarrow \cos x \, dx = du$

$$\int \sec x \, dx = \int \frac{1}{\cos x} dx = \int \frac{\cos x}{1-\sin^2 x} dx = \int \frac{du}{1-u^2} = \frac{1}{2}\int \frac{du}{1-u} + \frac{1}{2}\int \frac{du}{1+u}$$

$$= -\frac{1}{2} \ln|1-u| + \frac{1}{2} \ln|1+u| + c$$

$$= \ln\sqrt{\frac{1+u}{1-u}} + c = \ln\sqrt{\frac{1+\sin x}{1-\sin x}} + c = \ln\sqrt{\frac{(1+\sin x)^2}{\cos^2 x}} + c$$

$$= \ln\left|\frac{1+\sin x}{\cos x}\right| + c = \ln|\sec x + \tan x| + c$$

(44) $$\int \frac{x^3+x}{x-1} dx = \int\left(x^2+x+2+\frac{2}{x-1}\right)dx = \frac{x^3}{3} + \frac{x^2}{2} + 2x + 2\ln|x-1| + c$$

$$\begin{array}{r|l} x^3+x & x-1 \\ \cline{2-2} -\ x^3-x^2 & x^2+x+2 \\ \hline x^2+x & \\ -\ x^2-x & \\ \hline 2x & \\ -\ 2x-2 & \\ \hline 2 & \end{array}$$

(45) $$\int \frac{x^2+2x-1}{2x^3+3x^2-2x} dx = A\int\frac{dx}{x} + \frac{B}{2}\int\frac{2dx}{2x-1} + C\int\frac{dx}{x+2}$$

$2x^3+3x^2-2x = x(2x-1)(x+2)$

$x^2+2x-1 = A(2x-1)(x+2) + Bx(x+2) + Cx(2x-1)$

$2A+B+2C = 1$

$3A+2B-C = 2$

$-2A = -1 \quad \Rightarrow A = 1/2 \quad , B = 1/5 \quad C = -1/10$

$$\int \frac{x^2+2x-1}{2x^3+3x^2-2x} dx = \frac{1}{2}\int\frac{dx}{x} + \frac{1}{10}\int\frac{2dx}{2x-1} - \frac{1}{10}\int\frac{dx}{x+2}$$

$$= \frac{1}{2}\ln|x| + \frac{1}{10}\ln|2x-1| - \frac{1}{10}\ln|x+2| + c$$

$$= \frac{1}{10}\ln\left|\frac{x^9(2x-1)}{x+2}\right| + c$$

(46) $$\int \frac{dx}{x^2-a^2} = \frac{1}{2a}\int \frac{dx}{x-a} - \frac{1}{2a}\int \frac{dx}{x+a}$$

$$= \frac{1}{2a} \ln|x-a| - \frac{1}{2a} \ln|x+a| + c = \frac{1}{2a} \ln\left|\frac{x-a}{x+a}\right| + c$$

(47) $$\int \frac{x^4-2x^2+4x+1}{x^3-x^2-x+1}\,dx = \int\left(x+1+\frac{4x}{x^3-x^2-x+1}\right)dx =$$

$$\begin{array}{r|l} \cancel{x^4}-2x^2+4x+1 & x^3-x^2-x+1 \\ \hline -/\ \cancel{x^4}-x^3-x^2+x & x+1 \\ \hline x^3-x^2+3x+1 & \\ -/\ x^3-x^2-x+1 & \\ \hline 4x & \end{array}$$

$$x^3-x^2-x+1 = x^2(x-1)-(x-1)$$
$$= (x-1)(x^2-1) = (x-1)^2(x+1)$$

$$\int \frac{x^4-2x^2+4x+1}{x^3-x^2-x+1} = \int (x+1)\,dx + A\int \underset{(x^2-1)}{\frac{dx}{x-1}} + B\int \underset{(x+1)}{\frac{dx}{(x-1)^2}} + C\int \underset{(x-1)^2}{\frac{dx}{x+1}}$$

$$4x = Ax^2-A+Bx+B+Cx^2-2Cx+C = (A+C)x^2+(B-2C)x-A+B+C$$

$$\left.\begin{array}{l} A+C=0 \\ B-2C=4 \\ -A+B+C=0 \end{array}\right\} \left.\begin{array}{l} B-2C=4 \\ B+2C=0 \end{array}\right\} \Rightarrow B=2 \quad C=-1 \quad A=1$$

$$\int \frac{x^4-2x^2+4x+1}{x^3-x^2-x+1} = \int (x+1)dx + \int\frac{dx}{x-1} + 2\int\frac{dx}{(x-1)^2} - \int\frac{dx}{x+1}$$

$$= \frac{x^2}{2} + x + \ln|x-1| - \frac{2}{x-1} - \ln|x+1| + c$$

$$= \frac{x^2}{2} + x - \frac{2}{x-1} + \ln\left|\frac{x-1}{x+1}\right| + c$$

(48) $$\int \frac{2x^2-x+4}{x^3+4x}\,dx = A\int\frac{dx}{x} + \int\frac{Bx+C}{x^2+4}$$

$$2x^2-x+4 = Ax^2+4A+Bx^2+Cx = (A+B)x^2+Cx+4A$$

$$A=1 \ , \quad C=-1 \ , \quad B=1$$

$$\int \frac{2x^2-x+4}{x^3+4x}\,dx = \int\frac{dx}{x} + \int\frac{x-1}{x^2+4}\,dx = \int\frac{dx}{x} + \frac{1}{2}\int\frac{2x\,dx}{x^2+4} - \int\frac{dx}{x^2+4}$$

$$= \ln x + \frac{1}{2}\ln(x^2+4) - \frac{1}{2}\arctan\frac{x}{2} + c$$

(49) $\int \frac{4x^2-3x+2}{4x^2-4x+3}dx = \int\left(1+\frac{x-1}{4x^2-4x+3}\right)dx = \int dx + \frac{1}{4}\int\frac{8x-4}{4x^2-4x+3}dx$

$$\begin{array}{r|l} 4x^2-3x+2 & 4x^2-4x+3 \\ -(4x^2-4x+3) & 1 \\ \hline x-1 & \end{array}$$

$4x^2-4x+3=(2x-1)^2+2$

$2x-1=\sqrt{2}\tan t$

$2dx=\sqrt{2}\sec^2 t\,dt$

$$\int\frac{4x^2-3x+2}{4x^2-4x+3}dx = \int dx + \frac{1}{8}\int\frac{(8x-4)dx}{4x^2-4x+3} + \frac{1}{8}\int\frac{-4dx}{4x^2-4x+3}$$

$$= x+\frac{1}{8}\ln(4x^2-4x+3) - \frac{1}{4}\int\frac{2dx}{(2x-1)^2+2}$$

$$= x+\frac{1}{8}\ln(4x^2-4x+3) - \frac{1}{4}\int\frac{\sqrt{2}\cancel{\sec^2 t}\,dt}{2\cancel{\sec^2 t}}$$

$$= x+\frac{1}{8}\ln(4x^2-4x+3) - \frac{1}{4}\cdot\frac{\sqrt{2}}{2}\int dt$$

$$= x+\frac{1}{8}\ln(4x^2-4x+3) - \frac{\sqrt{2}}{8}\arctan\left(\frac{\sqrt{2}}{2}(2x-1)\right)+C$$

(50) $\int\frac{1-x+2x^2-x^3}{x(x^2+1)^2}dx = A\int\frac{dx}{x}+\int\frac{Bx+C}{x^2+1}+\int\frac{Dx+E}{(x^2+1)^2}$

$A+B=0$ $\quad C=-1$

$2A+B+D=2$ $\quad C+E=-1$ $\quad A=1$ $\quad \Rightarrow \quad A=1 \quad B=-1 \quad C=-1 \quad D=1 \quad E=0$

$$\int\frac{1-x+2x^2-x^3}{x(x+1)^2}dx = \int\frac{dx}{x}+\int\frac{x+1}{x^2+1}dx+\int\frac{x\,dx}{(x^2+1)^2}$$

$$= \int\frac{dx}{x} - \frac{1}{2}\int\frac{2x\,dx}{x^2+1} - \int\frac{dx}{x^2+1} + \frac{1}{2}\int\frac{du}{u^2}$$

$$= \ln|x| - \frac{1}{2}\ln(x^2+1) - \arctan x - \frac{1}{2}\frac{1}{x^2+1}+C$$

(51) $\int \frac{\sqrt{x+4}}{x}\,dx$

$x+4=u^2 \Rightarrow x=u^2-4 \Rightarrow dx=2u\,du$

$x=u^2-4$

$$\int \frac{\sqrt{x+4}}{x}\,dx=\int \frac{u\cdot 2u\,du}{u^2-4}=2\int \frac{u^2}{u^2-4}\,du=2\int\left[1+\frac{4}{u^2-4}\right]du$$

$$=2\int du+8\int\frac{du}{u^2-4}=2u+8\left[\frac{1}{4}\int\frac{du}{u-2}-\frac{1}{4}\int\frac{du}{u+2}\right]$$

$$=2u+2\ln|u-2|-2\ln|u+2|+c$$

$$=2\sqrt{x-4}+2\ln\left|\frac{\sqrt{x-4}-2}{\sqrt{x-4}+2}\right|+c$$

(52) $\int \frac{\cos x\,dx}{\sin^2 x+2\sin x-3}=\int\frac{du}{u^2+2u-3}=A\int\frac{du}{u+3}+B\int\frac{du}{u-1}$

$\sin x=u \Rightarrow \cos x\,dx=du$

$1=(A+B)u+-A+3B$

$A+B=0$

$B=1/4 \Rightarrow A=-1/4$

$-A+3B=1$

$$\int\frac{\cos x\,dx}{\sin^2x+2\sin x-3}=\frac{1}{4}\int\frac{du}{u-1}-\frac{1}{4}\int\frac{du}{u+3}=\frac{1}{4}\ln\left|\frac{u-1}{u+3}\right|+c$$

$$=\frac{1}{4}\ln\left|\frac{\sin x-1}{\sin x+3}\right|+c$$

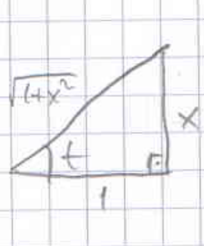
(53) $\int\frac{dx}{(1+x^2)^2}=$

$x=\tan t \Rightarrow dx=\sec^2 t\,dt \qquad (1+x^2)^2=(1+\tan^2 t)^2=(\sec^2 t)^2=\sec^4 t$

$$\int\frac{dx}{(1+x^2)^2}=\int\frac{\sec^2 t\,dt}{\sec^4 t}=\int\cos^2 t\,dt=\int\frac{1+\cos 2t}{2}\,dt$$

$$=\frac{1}{2}\int dt+\frac{1}{2}\int\cos 2t\,dt=\frac{t}{2}+\frac{1}{4}\underbrace{\sin 2t}_{2\sin t\cdot\cos t}+c$$

$$=\frac{1}{2}\arctan x+\frac{1}{4}\cdot 2\cdot\frac{x}{\sqrt{1+x^2}}\cdot\frac{1}{\sqrt{1+x^2}}+c$$



$$=\frac{1}{2}\arctan x+\frac{1}{2}\,\frac{x}{1+x^2}+c$$

II. Yol..

$$\int \frac{dx}{(1+x^2)^2} = \int \frac{1+x^2-x^2}{(1+x^2)^2}\,dx = \int \frac{dx}{1+x^2} - \int \frac{x^2dx}{(1+x^2)^2} = \arctan x - \int \frac{x^2dx}{(1+x^2)^2}$$

$x = u \Rightarrow dx = du \quad \frac{x}{(1+x^2)^2}dx = dv \,,\quad 1+x^2 = t \quad x\,dx = \frac{1}{2}dt$

$$v = \int dv = \int \frac{x\,dx}{(1+x^2)^2} = \frac{1}{2}\int \frac{dt}{t^2} = \frac{-1}{2t} = \frac{-1}{2(1+x^2)}$$

$$\int \frac{x^2dx}{(1+x^2)^2} = \int \underbrace{x}_{u} \cdot \underbrace{\frac{x\,dx}{(1+x^2)^2}}_{dv} = uv - \int v\,du = \frac{-x}{2(1+x^2)} + \frac{1}{2}\int \frac{dx}{1+x^2} = \frac{-x}{2(1+x^2)} + \frac{1}{2}\arctan x$$

Then;

$$\int \frac{dx}{(1+x^2)^2} = \arctan x - \left\{ \frac{-x}{2(1+x^2)} + \frac{1}{2}\arctan x \right\} + c = \frac{1}{2}\arctan x + \frac{1}{2}\,\frac{x}{1+x^2} + c$$

54 $$\int \frac{4x-3}{(x^2+1)^2}dx = \underbrace{\int \frac{4x}{(x^2+1)^2}dx}_{I_1} - 3\underbrace{\int \frac{dx}{(x^2+1)^2}}_{I_2}$$

$x^2+1 = t$
$2x\,dx = dt$

$$I_1 = 2\int \frac{2x\,dx}{(x^2+1)^2} = 2\int \frac{dt}{t^2} = -2\,\frac{1}{x^2+1} + C$$

$$\int \frac{4x-3}{(x^2+1)^2}dx = -\frac{2}{x^2+1} - \frac{3}{2(1+x^2)} - \frac{3}{2}\arctan x + c$$

55 $$\int \frac{1+\sin x}{1-\cos x}dx = \int \frac{1+\frac{2t}{1+t^2}}{1-\frac{1-t^2}{1+t^2}} \cdot \frac{2\,dt}{1+t^2} = \int \frac{t^2+2t+1}{\cancel{2}t^2} \cdot \frac{\cancel{2}dt}{1+t^2} = \int \frac{t^2+2t+1}{t^2(1+t^2)}dt$$

$\tan\frac{x}{2} = t \Rightarrow \sin x = \frac{2t}{1+t^2} \,,\quad \cos x = \frac{1-t^2}{1+t^2} \,,\quad dx = \frac{2dt}{1+t^2}$

$$\int \frac{1+\sin x}{1-\cos x}dx = \int \frac{t^2+2t+1}{t^2(1+t^2)}dt = \int \frac{t^2+1}{t^2(t^2+1)}dt + \int \frac{2t}{t^2(t^2+1)}dt = \int \frac{dt}{t^2} + \int \frac{2\,dt}{t(t^2+1)}$$

$$= \frac{-1}{t} + 2\int \frac{dt}{t} - \int \frac{2t}{t^2+1}dt = -\frac{1}{t} + 2\ln|t| - \ln|t^2+1| + c$$

$$= -\cot\frac{x}{2} + \ln\left(\frac{\tan^2(x/2)}{\sec^2(x/2)}\right) + c = -\cot\frac{x}{2} + \ln(\sin^2(x/2)) + c$$

(56) $\int \frac{dx}{2\cos x + 3\sin x + 4}$

$\tan\frac{x}{2} = t \Rightarrow \sin x = \frac{2t}{1+t^2}$, $\cos x = \frac{1-t^2}{1+t^2}$, $dx = \frac{2\,dt}{1+t^2}$

$$\int \frac{dx}{2\cos x + 3\sin x + 4} = \int \frac{\frac{2\,dt}{1+t^2}}{\frac{2-2t^2}{1+t^2} + \frac{6t}{1+t^2} + 4} = \int \frac{2\,dt}{2-2t^2+6t+4+4t^2}$$

$$= \int \frac{2\,dt}{2t^2+6t+6} = \int \frac{dt}{t^2+3t+3} = \int \frac{dt}{\left(t+\frac{3}{2}\right)^2 + \frac{3}{4}}$$

$t + \frac{3}{2} = \sqrt{\frac{3}{4}}\tan\theta$

$\left(t+\frac{3}{2}\right)^2 = \frac{3}{4}\tan^2\theta$

$dt = \sqrt{\frac{3}{4}}\sec^2\theta\, d\theta$

$$= \int \frac{\sqrt{\frac{3}{4}}\,\cancel{\sec^2\theta}}{\frac{3}{4}\,\cancel{(\tan^2\theta+1)}}\,d\theta = \sqrt{\frac{4}{3}}\int d\theta = \frac{2\sqrt{3}}{3}\theta + c$$

$$= \frac{2\sqrt{3}}{3}\arctan\left(\frac{2\sqrt{3}}{3}\left(t+\frac{3}{2}\right)\right) + c$$

$$= \frac{2\sqrt{3}}{3}\arctan\left[\frac{2\sqrt{3}}{3}\arctan\frac{x}{2} + \sqrt{3}\right] + c$$

(57) $\int \frac{\sqrt{x}}{1+\sqrt[4]{x^3}}\,dx$

$x^3 = u^4 \quad x = u^{4/3} \Rightarrow dx = \frac{4}{3}u^{1/3}\,du \quad x^{1/2} = u^{2/3}$

$$\int \frac{\sqrt{x}}{1+\sqrt[4]{x^3}}\,dx = \int \frac{u^{2/3}}{1+u}\cdot\frac{4}{3}u^{1/3}\,du = \frac{4}{3}\int \frac{u}{1+u}\,du = \frac{4}{3}\int\left(1 - \frac{1}{1+u}\right)du$$

$$= \frac{4u}{3} - \frac{4}{3}\ln|1+u| + c = \frac{4}{3}\sqrt[4]{x^3} - \frac{4}{3}\ln\left|1+\sqrt[4]{x^3}\right| + c$$

(58) $\int \frac{\sqrt{x+1}-1}{\sqrt{x+1}+1}\,dx$

$x+1 = u^2 \Rightarrow dx = 2u\,du$

$$\int \frac{\sqrt{x+1}-1}{\sqrt{x+1}+1}\,dx = \int \frac{u-1}{u+1}2u\,du = \int \frac{2u^2-2u}{u+1}\,du$$

$$\begin{array}{r|l} 2u^2-2u & u+1 \\ \hline -2u^2+2u & 2u-4 \\ -4u & \\ -/\ -4u-4 & \\ 4 & \end{array}$$

$$= \int\left(2u - 4 + \frac{4}{u+1}\right)du$$

$$= u^2 - 4u + 4\ln|u+1| + c = x+1-4\sqrt{x+1} + 4\ln\left|\sqrt{x+1}+1\right| + c$$

(59) $\displaystyle\int \frac{\sqrt[3]{x}+1}{3+\sqrt{x}}\,dx$

$$x = u^6 \Rightarrow dx = 6u^5\,du \qquad \sqrt[3]{x} = u^2 \qquad \sqrt{x} = u^3$$

$$\int \frac{\sqrt[3]{x}+1}{3+\sqrt{x}}\,dx = \int \frac{u^2+1}{u^3+3}\,6u^5\,du = 6\int \frac{u^7+u^5}{u^3+3}\,du$$

$$\begin{array}{l|l} u^7+u^5 & u^3+3 \\ \hline -/\ u^7+3u^4 & u^4+u^2-3u \\ \quad u^5-3u^4 & \\ -/\ u^5+3u^2 & \\ \quad -3u^4-3u^2 & \\ -/\ -3u^4-9u & \\ \quad -3u^2+9u & \end{array}$$

$$= 6\int \left(u^4+u^2-3u - \frac{3u^2-9u}{u^3+3} \right) du$$

$$= \int (6u^4+6u^2-18u)\,du - 6\int \frac{3u^2-9u}{u^3+3}\,du$$

$$= \frac{6}{5}u^5 + 2u^3 - 9u^2 - 18\underbrace{\int \frac{u^2-3u}{u^3+3}\,du}_{I}$$

$$3 = a^3 \Rightarrow I = \int \frac{u^2-3u}{u^3+a^3}\,du = \int \frac{A\,du}{u+a} + \int \frac{Bu+C}{u^2-ua+a^2}$$

$$u^2-3u = A(u^2-ua+a^2) + (Bu+C)\cdot(u+a) = A(u^2-ua+a^2) + Bu^2 + (aB+C)u + aC$$

$$= \underbrace{(A+B)}_{1}u^2 + \underbrace{(aB+C-aA)}_{3}u + \underbrace{aC+a^2A}_{0}$$

$$A+B = 1 \Rightarrow B = 1-A$$

$$aB+C-aA = a(1-A)+C-aA \overset{a/}{\Rightarrow} 2aA - C = a$$

$$\begin{array}{rl} & 2aA - C = a \\ + & a^2A + aC = 0 \\ \hline & 3a^2A = a^2 \end{array} \qquad A = 1/3 \Rightarrow B = 2/3$$

$$C = \frac{2a}{3} - a = -\frac{1}{3}a$$

$$I = \frac{1}{3}\int \frac{du}{u+a} + \frac{1}{3}\int \frac{2u-a}{u^2-ua+a^2}\,du = \frac{1}{3}\ln|u+a| + \frac{1}{3}\ln(u^2-ua+a^2)$$

$$\Rightarrow \int \frac{\sqrt[3]{x}+1}{3+\sqrt{x}}\,dx = \frac{6}{5}u^5 + 2u^3 - 9u^2 - 18\left[\frac{1}{3}\ln|u+a| + \frac{1}{3}\ln(u^2-ua+a^2)\right] + c$$

$$= \frac{6}{5}x^{5/6} + 2\sqrt{x} - 9\sqrt[3]{x} - 6\ln\left|\sqrt[6]{x} + \sqrt[3]{3}\right| - 6\ln\left(\sqrt[3]{x} - \sqrt[6]{9x} + \sqrt[3]{9}\right) + c$$

(60) $\displaystyle\int \frac{x+1}{(x^2-x+1)^2}\,dx$

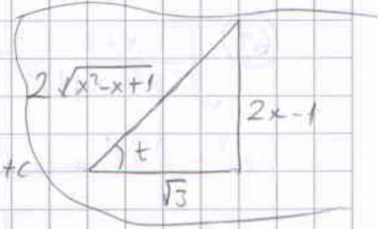
$x-\frac{1}{2}=\frac{\sqrt{3}}{2}\tan t \Rightarrow x^2-x+\frac{1}{4}=\frac{3}{4}\tan^2 t \Rightarrow x^2-x+1=\frac{3}{4}(\tan^2 t+1)=\frac{3}{4}\sec^2 t$

$dx=\frac{\sqrt{3}}{2}\sec^2 t\,dt$

$\displaystyle\int \frac{\left(\frac{\sqrt{3}}{2}\tan t+\frac{1}{2}\right)}{\frac{9}{16}\cancel{\sec^4 t}_{\sec^2 t}}\frac{\sqrt{3}}{2}\cancel{\sec^2 t}\,dt=\frac{4}{9}\int\frac{3\tan t+\sqrt{3}}{\sec^2 t}\,dt=\frac{4}{9}\int\left(3\sin t\cos t+\sqrt{3}\cos^2 t\right)dt$

$\displaystyle=\frac{\overset{2}{\cancel{6}}}{\underset{3}{\cancel{9}}}\int \sin 2t\,dt+\frac{\overset{2}{\cancel{4}}\sqrt{3}}{9}\int\frac{1+\cos 2t}{\cancel{2}}\,dt=\frac{2}{3}\cdot\frac{1}{\cancel{2}}(-\cos 2t)+\frac{2\sqrt{3}}{9}t+\frac{2\sqrt{3}}{9}\cdot\frac{1}{\cancel{2}}\sin 2t+c$

$=\frac{\sqrt{3}}{9}\sin 2t-\frac{1}{3}\cos 2t+\frac{2\sqrt{3}}{9}t+c$



$=\frac{\sqrt{3}}{9}\cdot\cancel{2}\cdot\frac{2x-1}{\cancel{2}\sqrt{x^2-x+1}}\cdot\frac{\sqrt{3}}{2\sqrt{x^2-x+1}}-\frac{1}{3}\left(\frac{3-(2x-1)^2}{4(x^2-x+1)}\right)+\frac{2\sqrt{3}}{9}\arctan\left(\frac{2x-1}{\sqrt{3}}\right)+c$

$=\frac{2x-1}{6(x^2-x+1)}-\frac{3-4x^2+4x-1}{12(x^2-x+1)}+\frac{2\sqrt{3}}{9}\arctan\left(\frac{2x-1}{\sqrt{3}}\right)+C$

II.61.

(61) $\displaystyle\int \frac{dx}{\sin^2 x} = -\cot x + C$

(62) $\displaystyle\int \frac{7}{(3x+2)^n}\,dx$

$u = 3x+n \Rightarrow dx = \dfrac{du}{3}$

$$\int \frac{7\,dx}{(3x+2)^n} = \frac{7}{3}\int \frac{du}{u^n} = \frac{7}{3}\,\frac{u^{-n+1}}{-n+1} + C = \frac{7}{3(1-n)\,(3x+n)^{n-1}}$$

(63) $\displaystyle\int \frac{x+2}{(x^2-x+1)^2}\,dx$

$x^2 - x + 1 = u \Rightarrow (2x-1)\,dx = du$

$$\int \frac{x+2}{(x^2-x+1)^2} = \frac{1}{2}\int \frac{2x-1}{(x^2-x+1)^2}\,dx + \frac{5}{2}\int \frac{dx}{(x^2-x+1)^2} = \frac{1}{2}\int \frac{du}{u^2} + \frac{5}{2}\int \frac{dx}{(x^2-x+1)^2}$$

$$= \frac{-1}{2(x^2-x+1)} + \frac{5}{2}\int \frac{dx}{\left[(x-1/2)^2 + \frac{3}{4}\right]^2}$$

$x - \dfrac{1}{2} = \dfrac{\sqrt{3}}{2}\tan t$

$dx = \dfrac{\sqrt{3}}{2}\sec^2 t$

$$= \frac{-1}{2(x^2-x+1)} + \frac{5}{2}\int \frac{\frac{\sqrt{3}}{2}\sec^2 t\,dt}{\frac{9}{16}\sec^4 t}$$

$$= \frac{-1}{2(x^2-x+1)} + \frac{20\sqrt{3}}{9}\int \cos^2 t\,dt = \frac{-1}{2(x^2-x+1)} + \frac{20\sqrt{3}}{9}\left(\frac{t}{2} + \frac{\sin 2t}{4}\right) + C$$

$$= \frac{-1}{2(x^2-x+1)} + \frac{10\sqrt{3}}{9}\arctan\left(\frac{2x-1}{\sqrt{3}}\right) + \frac{5\sqrt{3}}{9}\cancel{2}\,\frac{\sqrt{3}}{\cancel{2}\sqrt{x^2-x+1}}\cdot\frac{2x-1}{2\sqrt{x^2-x+1}}$$

$$= \frac{10x-8}{6(x^2-x+1)} + \frac{10\sqrt{3}}{9}\arctan\left(\frac{2x-1}{\sqrt{3}}\right) + C$$

(64) $\int \frac{dx}{(x+2)(x-1)^2}$

$$\frac{1}{(x+2)(x-1)^2} = \frac{A}{x+2} + \frac{B}{x-1} + \frac{C}{(x-1)^2}$$

$$1 = A(x^2-2x+1) + B(x^2+x-2) + C(x+1) = (A+B)x^2 + (-2A+B+C)x + A-2B+2C$$

$$\begin{cases} A+B=0 \\ -2A+B+C=0 \\ A-2B+2C=1 \end{cases} \Rightarrow \left[\begin{array}{ccc|c} 1 & 1 & 0 & 0 \\ -2 & 1 & 1 & 0 \\ 1 & -2 & 2 & 1 \end{array}\right] \sim \left[\begin{array}{ccc|c} 1 & 1 & 0 & 0 \\ 0 & 3 & 1 & 0 \\ 0 & -3 & 2 & 1 \end{array}\right] \sim \left[\begin{array}{ccc|c} 1 & 1 & 0 & 0 \\ 0 & 1 & 1/3 & 0 \\ 0 & 0 & 3 & 1 \end{array}\right]$$

$$\sim \left[\begin{array}{ccc|c} 1 & 1 & 0 & 0 \\ 0 & 1 & 0 & -1/9 \\ 0 & 0 & 1 & 1/3 \end{array}\right] \sim \left[\begin{array}{ccc|c} 1 & 0 & 0 & 1/9 \\ 0 & 1 & 0 & -1/9 \\ 0 & 0 & 1 & 1/3 \end{array}\right]$$

$$\int \frac{dx}{(x+2)(x-1)^2} = \frac{1}{9}\int \frac{dx}{x+2} - \frac{1}{9}\int \frac{dx}{x-1} + \frac{1}{3}\int \frac{dx}{(x-1)^2}$$

$$= \frac{1}{9} \ln\left|\frac{x+2}{x-1}\right| - \frac{1}{3(x-1)} + C$$

(65) $\int \frac{e^{\arctan x}}{1+x^2} dx$

$$u = \arctan x \Rightarrow du = \frac{dx}{1+x^2}$$

$$\int \frac{e^{\arctan x}}{1+x^2} dx = \int e^u du = e^u + C = e^{\arctan x} + C$$